

COL DU GRAND ST BERNARD, Switzerland

WMO: 067170

Lat: 45.8689N

Lon: 7.1708E

Elev: 2479

StdP: 74.88

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-18.1	-16.3	-28.4	0.4	-9.2	-26.5	0.5	-9.2	18.4	-7.6	16.3	-8.3	7.1	90	0.900

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.2	16.7	9.4	15.3	9.0	14.0	8.6	11.0	14.2	10.2	13.4	9.5	12.7	4.6	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.7	10.2	11.8	8.9	9.6	11.3	8.0	9.0	10.7	38.9	14.1	36.7	13.4	34.8	12.8	13.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.5	13.1	11.4	DB	-20.0	18.6	2.3	1.4	-21.7	19.6	-23.0	20.5	-24.3	21.2	-26.0	22.3	(4)
(5)			WB	-20.4	12.2	2.1	0.4	-21.9	12.5	-23.2	12.8	-24.4	13.0	-26.0	13.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	0.1	-7.3	-7.9	-5.3	-2.1	1.3	6.8	8.7	8.5	5.0	1.8	-3.4	-5.9
	DBStd	7.14	4.30	4.70	4.19	3.54	3.68	4.03	3.54	3.69	3.67	4.09	4.39	4.26
	HDD10.0	3695	536	500	476	364	269	112	65	71	154	253	402	494
	HDD18.3	6667	794	733	734	614	527	347	298	303	401	511	652	752
	CDD10.0	68	0	0	0	0	0	15	25	26	2	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	5.3	5.5	5.6	5.6	5.5	5.4	4.8	4.8	4.7	5.1	5.3	5.7	5.5
Precipitation	PrecAvg	1210	107	99	86	97	111	99	101	95	82	93	125	116
	PrecMax	1656	265	282	266	160	225	199	185	183	192	243	383	377
	PrecMin	910	18	13	13	17	36	28	42	33	19	18	33	0
	PrecStd	166	74	74	60	44	57	41	36	38	41	53	81	83
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.5	4.4	4.4	7.0	12.2	18.2	18.6	18.3	14.2	11.4	8.1	3.4
		MCWB	-2.2	-2.3	-1.1	1.7	6.4	9.0	9.8	9.5	7.8	5.0	1.0	-2.5
	2%	DB	1.6	2.2	2.8	5.4	9.9	16.0	16.4	16.6	12.6	9.5	5.2	1.7
		MCWB	-3.1	-3.3	-2.1	1.1	5.5	8.8	9.4	9.8	7.2	3.9	0.2	-3.1
	5%	DB	0.2	0.2	1.4	4.1	8.1	14.3	15.0	15.1	11.2	8.2	3.7	0.6
		MCWB	-3.4	-3.8	-2.5	0.7	4.5	8.3	9.0	9.1	6.7	3.3	-0.4	-3.4
	10%	DB	-1.2	-1.6	0.1	2.7	6.5	12.6	13.7	13.6	9.8	6.8	2.2	-0.4
		MCWB	-3.9	-4.4	-2.9	0.0	3.5	8.0	8.7	8.6	6.0	2.6	-1.2	-3.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.1	-0.9	0.3	3.0	7.2	11.1	12.0	11.8	9.6	6.6	2.7	-0.1
		MCDB	0.8	1.1	2.1	4.9	9.9	14.5	14.8	14.9	12.0	8.9	4.7	0.9
	2%	WB	-1.3	-2.0	-0.8	1.9	6.1	10.1	11.0	11.0	8.5	5.6	1.5	-1.2
		MCDB	-0.1	0.4	0.9	4.2	9.2	13.4	14.0	14.4	10.9	7.6	3.9	0.2
	5%	WB	-2.6	-3.1	-1.5	1.2	5.0	9.4	10.2	10.1	7.6	4.8	0.3	-2.2
		MCDB	-0.7	-0.1	0.2	3.6	7.6	12.9	13.4	13.3	10.0	6.7	2.4	-0.8
	10%	WB	-3.9	-4.4	-2.3	0.3	3.8	8.4	9.5	9.4	6.8	4.0	-0.5	-3.2
		MCDB	-1.5	-1.8	-0.5	2.4	5.8	12.0	12.7	12.5	9.1	5.8	1.2	-1.5
Mean Daily Temperature Range	5% DB	MDBR	4.5	4.5	4.6	4.5	4.8	5.7	6.2	5.8	5.2	4.6	4.1	4.5
		MCDBR	5.4	5.7	5.4	5.4	6.3	7.1	7.3	6.8	6.4	5.8	4.8	5.0
		MCWBR	4.3	4.0	3.7	3.6	3.5	3.9	4.0	4.0	3.9	4.1	3.2	3.5
	5% WB	MCDBR	5.4	5.6	4.7	4.8	5.9	6.6	6.7	6.3	5.7	5.0	4.3	4.8
		MCWBR	5.1	4.6	3.8	3.4	3.5	3.9	4.1	3.9	3.6	3.9	3.3	4.5
Clear-Sky Solar Irradiance	taub	0.203	0.220	0.252	0.284	0.297	0.302	0.298	0.290	0.276	0.259	0.229	0.204	
	taud	2.396	2.359	2.335	2.386	2.444	2.464	2.470	2.505	2.539	2.572	2.533	2.439	
	Ebn at Noon	951	988	990	979	970	963	962	961	953	931	911	916	
	Edh at Noon	68	87	106	111	109	108	106	98	86	71	60	58	
All-Sky Solar Radiation	RadAvg	1.55	2.41	3.51	4.40	5.03	6.00	6.06	5.28	4.11	2.67	1.56	1.28	
	RadStd	0.18	0.31	0.46	0.56	0.43	0.49	0.47	0.39	0.33	0.30	0.23	0.18	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (87 neighbors)		N/A	N/A	N/A	+0.68	+0.37	+0.26	N/A	N/A	+56	+18

Nomenclature: See separate page